

JUNGWOO KIM

Dept. of Electrical and Electronic Engineering, Yonsei University

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EDUCATION

Yonsei University Seoul, Korea
B.S. in Electrical and Electronic Engineering
GPA: 3.86 / 4.3 (Major: 3.95 / 4.3)

Mar. 2022 - Expected Feb. 2028

INDEPENDENT PROJECTS

Tactile-Feedback Robotic Hand

2026

- Built a two-modality tactile front-end for a robotic hand: FSR normal-force sensing through a 16-bit I2C ADC, and PVDF-film slip-vibration sensing through a mid-rail-biased $\times 51$ amplifier stage.
- Sized the amplifier gain by characterizing the sensor noise floor with an FFT on MCU-sampled data, after measuring that the unamplified slip signal fell below the ADC quantization floor
- Detected slip on the bench from the total variation of the signal over a 0.5 s window sampled at 5 kHz, separating idle from sliding contact by a factor of 1.7.
- Ported an open-source MediaPipe/MuJoCo/Mink teleoperation pipeline (dora-rs) to Windows for an 8-DOF robotic hand, tuning IK damping and update rate for stable tracking.

RESEARCH EXPERIENCE

RoPhi Lab, Yonsei University

Sep. 2026 - Present

Undergraduate Research Trainee

- Completing prerequisite training in robot kinematics, dynamics, and control using Modern Robotics before progressing to lab simulation and hardware training.

ACADEMIC PROJECTS

Line-Tracing Robot

2026

- Built an IR-sensor line-following robot on an Arduino Nano with a TB6612FNG motor driver, and tuned a PD controller for stable line tracking.

Wideband Fully Differential Amplifier

2026

- Designed a two-stage fully differential amplifier in PSpice under fixed constraints (5 V single supply, one ideal voltage and one ideal current source, 300 Ω input, 1 pF load), meeting the gain and bandwidth targets.
- Applied a regulated cascode to the tail current source and repurposed the input and output transformer turns ratios as design variables, reaching 38.63 dB gain, 120.6 MHz bandwidth, and 247.92 dB CMRR at 14.6 mW.

Three-Band Audio Equalizer

2025

- Designed a three-stage equalizer with independent gain control for each band under a five-MOSFET constraint.
- Reached 24.27-24.93 dB in-band gain over the required 100 kHz bandwidth against a 4 dB minimum, and held in-band flatness to 0.66 dB against a 1 dB requirement.

Closed-Loop Variable-Speed DC Motor Drive Simulation

2026

- Simulated the dynamic performance of a closed-loop variable-speed drive for a 20 HP, 250 V, 600 rpm, 8-pole separately excited DC motor under a fan-type load.

WORK EXPERIENCE

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Jan. 2026 - Feb. 2026

Business Strategy Intern

- Proposed business direction based on market analysis and customer-needs research, through to project delivery.

Republic of Korea Air Force

Sep. 2023 - Jun. 2025

Sergeant

- Compulsory military service as required by the regulations of the government of the Republic of Korea.
- Performed maintenance and functional testing of data link pods (DLP) for the F-4E Phantom II fleet.

LEADERSHIP AND ACTIVITIES

Student Council, Dept. of Electrical and Electronic Engineering, Yonsei University

2023

Samsung Electronics Shining Star — semiconductor training program for undergraduates

2023

Entrepreneurship Society — emergency-vehicle alert system; presented at demo day

2025

SKILLS

Python, C/C++ (Arduino / ESP32), MATLAB, PSpice